# A Tinkerer's Mindset: Lessons from the Technical Leadership Institute's Co-Hort 8 on Safe-to-Fail Probing as a Tool for Informing Judgement

Adam D. Williams (Sandia National Laboratories*, Albuquerque, NM adwilli@sandia.gov); Leandro V. Aveiro (Alstom, Penfield, NY ); Rachel A. McGrath (Pratt &Whitney, East Hartford, CT); Carlo Leandri (Tetra Pak Packaging Solutions, Modena, Italy); Guilluame Terpant (Dassault Systems, Villacoublay, France); Dimitri Masson (University of Bourdeaux, France); Adrian Unger (VENG, Buenos Aires, Argentina)

**Abstract**. Tinkering—or making small changes to experiment toward an improvement in performance—is seemingly a natural characteristic of many systems engineers. As such, systems engineers are uniquely qualified to develop complex solutions necessary to overcome lack of clarity, achieve order, and avoid failure. Further, there is a much broader conversation surrounding the possibility of "failure" being beneficial in systems engineering projects. In response to the needing to inform judgment in situations shrouded in uncertainty, members of INCOSE's Technical Leadership Institute (TLI) Cohort 8 examined the role of safe-to-fail probes play in informing judgement for systems engineers. Within the constraints of the TLI's major project, virtual workshops and qualitative interviews were two data collection mechanisms established to empirically investigate the role(s) of safe-to-fail probing in systems engineering. Overall, the data sets offered conclusions describing the potential role(s) of safe-to-fail probes for systems engineers working in uncertain environments. Resulting from this (limited) empirical exploration are additional insights and implications for how systems engineers may invoke safe-to-fail probes to improve decision-making in uncertain and challenging situations. Such a Tinkerer's Mindset can help systems engineers transition from the constraints of "intolerable failure" to the opportunities related to probing-sensing-responding to "responsible failures."

## Background & Motivation

INCOSE's Technical Leadership Institute (TLI) seeks to build a global learning network of active INCOSE members seeking to improve their leadership skills in an open, collaborative environment. By engaging in an (initial) two-year experience composed of virtual/in-person workshops, coaching, collaborative projects, and asynchronous assignments, TLI participants gain knowledge and experience to become systems engineering leaders in the face of ambiguity and uncertainty. To date, the TLI has provided nearly 150 participants from six continents and approximately 20 countries with a tailored, rigorous, and collaborative program to support the development—and evolution—of systems engineers as *leaders* in addressing complex problems.

As an intentional and strategic effort by INCOSE to apply the unique skillset of the system engineer to intersection of "technical excellence" and "leaderships skills," TLI assigned Group 3 from Cohort 8 (aka, "The Warm Skeptics") to examine how to inform judgement in situations shrouded in various types of uncertainty. Related discussions within TLI/Cohort 8 navigated how the uncertain and the unknown can delay and deteriorate the relationship between judgement (or, evaluating a situation toward an opinion) and decision (or, making a choice). More specifically, Group 3 was assigned a 6-month long major project to examine the role of safe-to-fail probes play in informing judgement to deliver effective systems engineering decisions.

According to some systems engineering experts, current interpretations of the word "fail" is potentially problematic. For example, a 2010 panel of systems engineering experts asserted that

> *Society has been developing a hubris that we know how to do everything and that, every time we make a mistake, it is something we should have known better about…There is a perception in society that demands success the first time and every time. In contrast to*

> *perception, it is not possible to always avoid failure; therefore, it is vital to think and talk about what it means to learn from failure. (Slegers, et. al, 2012)*

The experts on this panel continued to review several major cases in which different types of failure occurred, ultimately concluding that only is failure an option at every system lifecycle step (until the final goal), failure *can provide* opportunities to reassess and improve (Slegers, et. al, 2012). Perhaps more succinctly, Elon Musk is often quoted as saying: "If things are not failing, you are not innovating enough" (Satara, 2018).

Though just a few anecdotes, this TLI group found that there is a much broader conversation surrounding the possibility of "failure" being *beneficial* in systems engineering projects. Despite attempts to manifest order, structure and coordination, systems engineers experience failures resulting from uncertain situations, indeterminate environments, or unknown scenarios—or a combination of all three. In response to suboptimal system performance, systems engineers often begin to make small changes to affect a repair or improvement in system performance—or tinker. Here, systems engineers naturally tinker to navigate a range of potential solutions necessary to overcome lack of clarity, achieve order, and avoid failure in complex problems. Though often seen as a difficulty (particularly in terms of decision-making), a systems engineers' tinkering can help reframe failures as opportunities to understand complexity. Such a Tinkerer's Mindset can help systems engineers transform a focus on avoiding failure to learning from controlled failures to navigate complexity.

After introducing safe-to-fail probes, situating them among leadership and complexity concepts, and mapping them to decision-making in complex and uncertain operational environments, this paper will describe two parallel—but related—data collection opportunities. Next, this paper will evaluate these empirical data to identify common themes and interesting trends describing how safe-to-fail probing may inform judgement to successfully execute systems engineering solutions. Lastly, this paper will review conclusions and offer insights from this empirical exploration, as well as suggest implications for how systems engineers may invoke safe-to-fail probes—similar to a Tinkerer's Mindset—to improve decision-making in uncertain and challenging situations.

## "Safe-to-Fail": Introduction to the Concept

"Safe-to-fail" refers to actions taken when attempting to make sense of complex situations or systems that result in suboptimal performance. Considering its broadest meaning, this study used the term "system" equally for engineered and natural systems—where both can produce uncertain outputs based on interactions between environmental and internal components. The potential for such uncertain outputs accentuates the challenge for systems engineers to make decisions in uncertain situations, indeterminate environments, or unknown scenarios. Yet, several approaches to informing judgement and sense-making related to complexity exist. Consider, for example, Snowden's Cynefin Framework (Cynefin.co, 2023) that was introduced the TLI's Cohort 8 early in their program. This framework has been employed successfully across many professional disciplines and routinely enhances the ability for decision-makers to evaluate their challenges, opportunities, and decisions in the correct context of their situation. More specifically, the Cynefin framework (Figure 1) categorizes systems (and situations) into five distinct groups (Cynefin.co, 2023):

- **Simple**: There is a linear and clear relationship between cause and effect and the relation is identifiable by most observers
- **Complicated:** There is a linear relationship between cause and effect, though that relation is not obvious and is likely only identifiable with proper application of expertise and analysis effort (Note: Reductionist approaches and traditional scientific methods are most suitable here)
- **Complex**: The relationship between cause and effect can only be observed in hindsight, as the number of interactions among internal elements and between the system and its environment creates emergent properties causing low or absent levels of predictability

- **Chaotic**: There is no relationship between cause and effect, as the interactions among internal elements and between the system and its environment results in absent levels of predictability or explainability
- **Disorder**: This is the state of not knowing where the system fits in, sometimes referred to as a state of ignorance

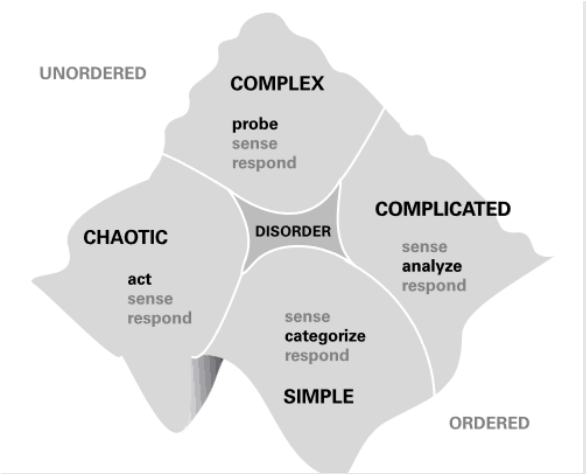



Figure 1. Original visual representation of Snowden's *Cynefin Framework* (Snowden and Boone, 2007)

For each of identified quadrants, the framework offers a pathway for sense making and informing judgment. For example, sense making for the complex quadrant is through *probe-sense-respond* cycles. In complex systems, there will be unintended consequences flowing from interactions in the systems. (Snowden, et. al, 2022). To that end, safe-to-fail experiments are targeted probes with the objective of minimizing any unintended outcomes toward making sense of complex behaviors. In other words, they are small-scale "experiments" designed from different perspectives with the intent to gain visibility on emergent properties of the system without significant damage (National Health Service, 2021).

While the Cynefin Framework offers probing as a first sense-making step for complexity, safe-to-fail experiments have some specific characteristics. First, they are not random attempts to stimulate the system and are targeted toward a set of learning objectives. Second, safe-to-fail probes need to be executed with caution and awareness of unacceptable outcomes. Lastly, the outcomes of safe-to-fail probes need to initiate additional actions—with a focus on recovering from a "failure" and amplifying a "success" (National Health Service, 2021).

Safe-to-fail experiments provide a method to identify the unknown "unknowns"—including poor decision-making, unnecessary rework, and (potential) mission failure that can result from incorrect sense-making. For instance, performing reductionist analysis—what is appropriate for the "complicated" Cynefin quadrant—on a complex system or situation as a mechanism for mitigating uncertainty can be quite costly and produce results with poor reliability. Yet, better understanding the roles and implications of probing into the unknown—even when what results could be classified as a failure—was the focus of this study.

## Data Collection to Explore the Concept

In response, this TLI major project required each group to collect data within the constraints of (1) drawing from fellow systems engineers (and, in some cases, only active and former TLI members) and (2) some form of interaction. Group 3 decided to leverage (and extend) initial interviews suggested by TLI for one data set. The interviews targeted business and technical leaders within (or adjacent to) INCOSE and

characterized the current status of safe-to-fail probes across a broader array of systems engineers. The second data emerged from virtual workshops conducted with members of the INCOSE TLI community. The workshops provided a natural experiment by which to evaluate the potential utility of safe-to-fail probes in future systems engineering applications. Both data collection mechanisms were established to empirically investigate the role(s) of safe-to-fail probing to inform judgement in systems engineering within the assignment description set by TLI and operational constraints set by Group 3 members.

## *Description of Interview Data*

Data set I consisted of qualitative interviews (Figure 2) because of their ability to support dynamic interaction, provide additional insight, capture nuance, and further explore unexpected responses (Weiss, 1995). Following a semi-structured process, these qualitative interviews provided for deeper descriptions of interviewees' true beliefs and interpretations of *how* and *why* failure is (mis)understood. Though not a traditional academic set of interviews, this qualitative interview data provides additional coherence, depth, and clarity with which to evaluate the role(s) of safe-to-fail probing. The team developed on a common set of guiding questions to be used during the interviews, which included aims to learn:

1) The interviewee's type of organization, domain, and professional background
2) Examples of interviewee making decisions in the face of complex scenarios
3) What decisions needed to be made and how the interviewee arrived at the conclusions
4) The difference between the expected and actual results
5) Any changes the interviewer would have made their approach in hindsight
6) The interviewee's (and organization's) perception of failure and when failure was accepted
7) If the interviewer could provide an example of a safe-to-fail probe that could have helped

Prior to question 2), each interviewer presented a description of the Cynefin framework that focused on the difference between the four quadrants but omitting the pathways for sense making and informing judgment. More specifically, a version of the Cynefin Framework diagram (Figure 1) was presented without the action words (e.g., "Complex" quadrant did not have the words "probe", "sense", "respond"). This characteristic of the interview design was aimed at maintaining authenticity and reducing interviewer bias toward probes if that was not the actual decision-making technique employed by the interviewee. Toward the end of each interview, prior to question 7), the interviewer then presented the complete description of the Cynefin Framework, now including the words "probe", "sense", and "respond" for the Complex quadrant.

| Int. | Domain | Current Role | ~ Years of Experience | INCOSE Involvement |
|---|---|---|---|---|
| A | Academia | Consultant | 50+ | Y<br>Working Group Leader |
| B | Defense[G,R] | Line/Program Manager | 15 | Y<br>Member |
| C | Defense[G,R] | Director | 30 | N |
| D | Aerospace | Major Projects Manager | 15 | N |
| E | Aerospace | General Manager | 23 | N |
| F | Food Packaging[C] | Technical Product Owner | 26 | N |
| G | Aerospace[G,C,R] | Assembly, Integration & Verification Lead Engineer | 25 | Y<br>Sector Director |
| H | Food Packaging[C] | Specialist Food Packaging | 30+ | N |
| I | Software, Academia | CEO/President & University Professor | 30+ | Y<br>Local Board Member |

| | | | | |
|---|---|---|---|---|
| J | Food Packaging[C] | Quality & Product Creation Director | 24 | N |
| K | Aerospace | Retired | 30+ | Y<br>Former Board Member |
| L | Defense | Vice President Systems Engineering Ecosystem | 30+ | Y<br>Member |
| M | Transport | Product Development Director | 20 | N |
| N | Imaging | Retired | 30+ | Y<br>Local Chapter Past President |
| [G] = government, [C] = commercial, [R] = research | | | | |

Figure 2. Demographic summary of interviews conducted to explore the "safe-to-fail" concept

Logistically, interviews lasted approximately one hour each. As possible, interviews were conducted in person; others were conducted virtually via Microsoft Teams or Zoom. Depending on availability of the technology, interviewees were asked for verbal consent for recording, note-taking or transcribing the meeting. Each interviewer then wrote a summary of their interviews, organizing the "raw data" according to the guiding questions, and shared them with other Group 3 members. For this study, all names of people, organizations, products and projects have been anonymized (Figure 2). In total, 14 interviews were included in this data set, including representation of systems engineering (or adjacent) professional roles from the academic, defense, transport, food packaging, and software domains. Approximately 50% of all interviewees are involved in INCOSE (to varying degrees), while all but two had more than 20 years of professional experience. While not a comprehensive grouping, these interviews do provide a representative sample of domains from which to draw useful conclusion and insights on safe-to-fail probing for systems engineering decision making.

## *Description of Workshop Data*

This data collection mechanism was designed to immerse participants in a hands-on experience in developing fail-to-safe probes for a complex scenario. The virtual workshop was conducted via Zoom, included virtual breakout rooms (for group-facilitated workshop activities), and utilized Miro as a virtual interactive space. The workshop was conducted in mid-October 2023, lasted 90 minutes, and was repeated in two different time-zones to support worldwide participation. Due to TLI-based constraints for this major project, participants consisted of members of from other TLI cohorts. The participants were split into two groups and the set of probes from each group were evaluated by the other. This mechanism simulated a complex scenario where responses are difficult to forecast and probes can play a relevant role in informing decisions in uncertain situations.

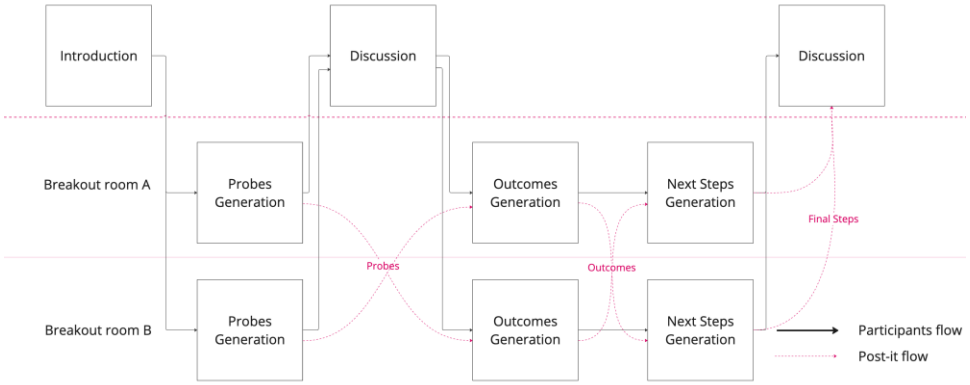



Figure 3. Conceptual design for the safe-to-fail workshop structure

The scenario at the heart of this data collection mechanism included the following characteristics:

- A main customer is claiming (perceived) safety issues on several production lots
- The suspect lots are partly delivered on the market and partly available in stock for quality checks
- Relationships with the customer is good but the long-standing customer manager recently retired
- The company specialist has engaged with legal, communications, and first alert technical personnel
- The customer is making a claim to take the road of harsh litigation
- Technical information is limited, non-structured and partly contradictory

In response, the participants were required to:

- Clarify the impact and the extension of the issue in terms of severity and actual occurrence
- Identify root causes and responsibilities so to eliminate the risk further occurrences
- Minimize the risk of losing customer confidence and avoid stakeholder reputational risks

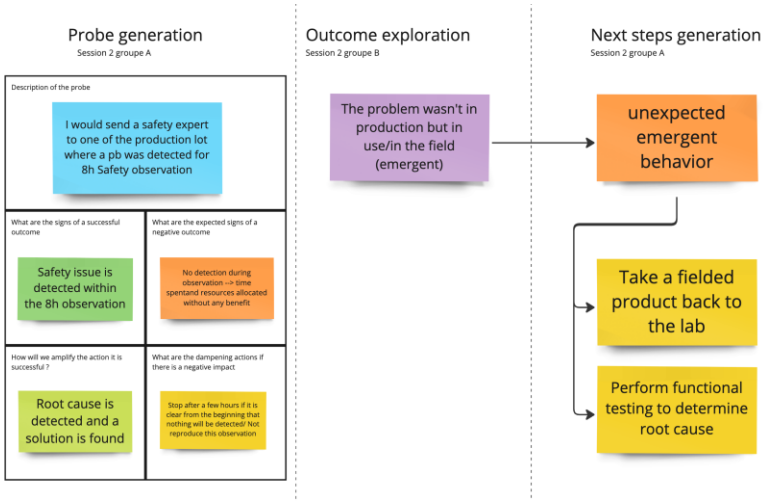



Figure 4. Example of the data collection from the virtual workshop on safe-to-fail probing.

Participants were provided with a structured template for probe generation on the Miro board. The template included a space for the probe prompt and four quadrants: signs of successful outcomes, expected signs of negative outcomes, strategies to amplify successful actions, and strategies to dampen negative outcomes. The workshop activities consisted of (Figure 3):

- *Probe generation*: individually and collaboratively create and produce safe-to-fail probes
- *Outcomes generation*: collaboratively generate possible outcomes to the prompts proposed by the opposite teams
- *Response generation*: Collaboratively compare outcomes to anticipated signs and generate final mitigation actions

| Session (Group) | Number of Participants | Number of probes [# in complex quadrant] | Number of outcomes | Number of next steps |
|---|---|---|---|---|
| 1(A) | 2 | 4 [2] | 10 | 8 |
| 1(B) | 3 | 7* [6] | 13 | 9 |
| 2(A) | 2 | 4 [3] | 5 | 7 |
| 2(B) | 2 | 4 [1] | 8 | 3 |
| *one had all four quadrants completed and two had only the prompt | | | | |

Figure 4. Summary of workshop results, including the number of probes derived and outcomes generated

The safe-to-fail probes generated by the participants were classified according to Cynefin framework quadrants by evaluating their use of good systems engineering practice for finding root causes (e.g., the complicated quadrant) or if the probes were more exploratory in nature (e.g., the complex quadrant). Per the workshop design, each probe elicited at least one response from the parallel team. The result was a rich diversity in the range of possible outcomes, particularly in terms of expectation and polarity. Some responses went *beyond* expectation and several responses used complexity to clearly reframe the problem space. Similarly, both positive and negative outcomes were generated in response to safe-to-fail probes. Lastly, each team prepared dampening and amplifying actions to address the probes and response. The workshop was a success in terms of creating the condition for participants to experiment with the concepts related to safe-to-fail probes. The generated outcomes allowed participants to revise their mindset and feel more comfortable exploring complexity and uncertainty.

## Data Analysis to Evaluate the Concept

Both data sets were iteratively analyzed with a focus on identifying trends, common themes, and areas for further exploration. Given the qualitative—and representative—nature of both data sets, the unit(s) of analysis were individual statements describing various aspects of probing, failure, complexity, uncertainty, and systems engineering. Here, analysis relied "less on counting and correlating and more on interpretation, summary and integration" (Weiss, 1995, p. 3) to investigate the potential role(s) of safe-to-fail probes for systems engineers.

### *Trends & Common Themes*

**Probing is common, safe-to-fail is not.** For the first common theme, perhaps unsurprisingly, emerging from the data sets is that *probing* and tinkering are a natural and regular occurrence in this representative set of systems engineers—but *safe-to-fail* experiments were not. This is reflected in the workshop by most probes stemming from common systems engineering problem finding methodologies than exploratory experimentation. Overall, 57% of the interviewees expressed that probing and experimenting is one of the

more immediate and obvious tools systems engineers use when dealing with complex systems. More specifically, consider the following examples:

- "I have a trial-and-error mindset" (Int A)
- "It's just how my brain works, I probe" (Int B)
- "Question everything, start prototyping per the scientific method." (Int C)
- "The key information derives from a combination of formal and standardized tests, focused on the interactions with the stakeholders …" (Int H)
- "Complex systems already naturally tend to show emergent behavior of amplitude difficult to be predicted. Daring to afford a potentially huge impact even if somehow controlled is not reasonable" (Int H)
- "Incremental validation of the emerging consensus …" (Int K)
- "We did try to allocate other resources, but that probe failed. After one or two weeks, it was clear they needed the experts on the problem." (Int M)

This is not an unexpected finding since the iterative process of experimentation and testing is commonly applied in systems engineering and scientific practices. Where probing was a natural reaction to exploring the uncertainty for a path forward, there was not a consensus among the interviewees on *how* to conduct such probing. Among our selected interviewees, the limits of each probe and their next steps seemed to be discovered spontaneously. In the workshops even when next steps were identified, participants usually differed in the final strategy when the outcomes were revealed. Conversely, none of the interviewees described more intentional probing strategies. For example, consider those exemplified in the Cynefin Framework, such as identifying tailored learning targets, defining acceptable failure levels, and implementing recovery (or amplification) strategies. Similarly, many of the probes proposed by the workshop participants stayed in the complicated quadrant—highlighting the natural tendency to resort to the complicated viewpoint rather than embrace the complexity of the situation (e.g., safe-to-fail probing). In this manner, one major theme in the data is that while probing may come naturally, safe-to-fail experiments do not.

**Common sense and intuition are still powerful tools.** A second trend emanating from the data relates to the roles of common sense and intuition for informing judgment and addressing uncertainty. Even though some situations described by our interviewees required probing to inform judgement, approximately 50% of the decisions evaluated were based on common sense and intuition built on past experiences. Again, this trend is consistent with the experience of many systems engineers, as human beings are "machines for jumping to conclusions" (Kahneman, 2011). This behavior is efficient if the conclusions are likely to be correct and the cost of an occasional mistake is acceptable. Jumping to conclusions is risky when the situation is unfamiliar, the stakes are high, and there is insufficient time to collect additional information. These are the circumstances in which intuitive errors are probable (Kahneman, 2011). Yet, both interview and workshop data illustrate that systems engineers are (roughly) equally likely to rely on past experiences—despite the possibile impacts of cognitive biases—and more formal analytic approaches to inform judgment under uncertainty. Consider the following statements as examples:

- "We didn't see the number of dumb things that were happening [resulting from intuition]" (Int. C)
- "[Arrived at the decisions] primary through discussions withs stakeholders achieving consensus on these ideas" (Int D)
- "Because there was a strong sense of duty tied to both ethical and moral considerations, I tried to listen to my conscience. I imagined myself from a distance, visualizing another person in my situation" (Int E)
- "The key information derives from a combination of formal and standardized tests … and the fifth sense deriving from long and recognized experience in the role" (Int H)
- "Confidence was the way to overcome complexity" (Int K)

- "... stand by the decision and even today I remain skeptical of whether the integration effort would indeed bring improvements" (Int M)
- "These [decisions] were all common sense to me." (Int N)

From the collected data, systems engineers are quick to invoke common sense and intuition. And, similar to the first common theme, there does seem to be a clear strategy, plan, or end game for navigating complexity and uncertainty. That only 50% of the interviewees indicated taking this approach suggests that there is an (at least equally) effective approach available for supporting strategic probing and a tinkerer's mindset.

**Failure seemingly has conflicting interpretations.** A final common theme emerging from the data related to perceptions of and connotations for the term "failure"—resulting in an overwhelming aversion to the term. Invariably, when asked about their opinions about failure, almost all interviewees agreed that controlled and measured failures are a consequence of learning and should be embraced. Whether failing under controlled, experimental situations (Int. D, Int. E) or as part of an intentional technical/design activity (Int. H, Int. M), interviewees did not default to being averse to failure. Yet, nearly all interviewees qualified these responses in terms of the perception of failure by their organizations. Overall, 43% of the interviewees reported their organizations would negatively view failures even in the context of informing judgement:

- "Perception is that failures will not be forgotten" (Int B)
- "Difference between 'safe to fail' and having the 'courage to fail' are executive leaders willing to be the 'bullet catcher'" (Int C)
- "Nevertheless, within the company, a mistake in a non-technical area is not well-received" (Int E)
- "A subcontractor is not expected to fail, just to execute what is required, on time" (Int F)
- "Although the processes encourage to dare potential limited failures (fail fast, fail safe), it is difficulty to break the glass ceiling of the historical 'jump to solution and fix it' practice" (Int I)
- "A lot can be learned by failure, but failing ends up getting a bad reputation" (Int N)

More specifically, interviewees further explained that tendency for failure aversion within their organizations stems from fear of safety impact (Int. H), reputational damage (Int. M), or financial loss (Int. D, Int. E). Yet, several interviewees stressed that "a lot can be learned by failure" (Int. N) and that "failure at the 'right' time...is part of the game" (Int. G). Thus, the aversion to failure seems to be a function of organizational culture and operational pressures around fears of unacceptable losses. Here, strategic and focused probing (like that proposed by the Cynefin Framework) offers opportunities to situate both interpretations of failure—namely by using targeted probes with well-defined and accepted levels of failures to explore uncertainty and learn complex solutions.

## *Areas for Additional Exploration*

To conclude each data collection exercise, three questions were posed to the participants to investigate opportunities to operationalize safe-to-fail probes for mitigating uncertainty and complexity, namely:

- Under what conditions—or with what characteristics—can safe-to-fail probes be successful?
- How can safe-to-fail probes be successfully implemented?
- What benefits can be anticipated with or experienced from safe-to-fail probes?

While there was not a clear consensus, the data revealed some candidate conditions and characteristics that support the successfully implementation of safe-to-fail probes. Some responses were explicit and clear, like Int. C's statement that safe-to-fail-based strategies can only be successful when provided with the necessary—and sufficient—authority, resources, and responsibility to adequately learn from shortcomings. Other characteristics were more philosophical, like Int. A' assertion that safe-to-fail probing requires stepping across the boundaries of what is known to work and Int. D's claim related to the psychological fortitude necessary to handle "failing."

Overall, safe-to-fail probes were deemed likely to be successful under two broad conditions. First, the associated "failure" does not involve safety, reputation, or financial losses. For example, in workshop session 2, one of the probes led to a catastrophic outcome, with participants noting that "[since the] probe has changed the situation [it was not a] safe-to-fail [experiment]". Second, the anticipated information or knowledge return seems worth the risk. Across both data sets, a positive interpretation of failure—similar to the basis of safe-to-fail probes—was associated with local actions or framed in terms of "traditional research."

The data also revealed additional considerations related to implementing safe-to-fail probes for systems engineering applications. One such consideration is offered by Int. C and Int. G, who both noted the need to gain stakeholder consensus and buy-in for defining acceptable failures. Int. M offered the ability to validate such consensus within a well-chosen core team—as well as the flexibility to revisit any such consensus as the stakeholder group grows—in implementing safe-to-fail probes. Lastly, Int. A suggested that need to "question everything" and "start prototyping" early and often, where Int. C asserted the need for a "did we learn from it" organizational and team paradigm to institute safe-to-fail probes.

Taken together, the conditions and actions that support safe-to-fail probes presuppose several benefits for decision-making in uncertain and complex situations. For example, an overwhelming majority of the statements supporting safe-to-fail probes highlighted the ability for systems engineers to explore a set of quasi-established, incomplete, or preliminary solutions to gauge which would provide the best next step. Stated more simply, safe-to-fail probes provide a mechanism for the systems engineer to learn from uncertain or complex situations without sacrificing substantial resources on more formal, detailed analysis. Such small, measured, and targeted probes can also assist in overcoming a full range of cognitive biases—despite the tendency for systems engineers to rely on "common sense" and intuition. Seemingly, the primary benefit of safe-to-fail probes is not to find why a situation is inadequate or uncertain, but rather to focus tinkering on where and how to apply novel recovery or amplification strategies to address complexity.

## Conclusions, Insights & Implications from the Concept

Overall, the data collected was sufficient to address the potential role(s) of safe-to-fail probes for systems engineers working in uncertain environments. Throughout both the interviews and workshops, the different interpretations of "failure" manifested an oxymoron, or an apparently contradictory figure of speech. More specifically, the data revealed that failure is only "bad" when nothing is learned—which suggests a logical consistency with the Cynefin "probe-sense-respond" approach where learning is a natural extension of responding. The data also illustrated that while failure can emerge in many different forms, the exploratory nature of safe-to-fail probing is a tool that can help identify drivers of uncertainty and complexity. And, one of the more consistent outcomes from the data was the realization that—particularly in complex situations—"failure is an inseparable part of learning" (Int. A). Though, likely not suitable in time-sensitive situations and decisions, the data suggests that systems engineers can invoke the "probe-sense-respond" ethos of safe-to-fail probing with planning, iteration, and time to explore the results.

Yet, the TLI requirements for data collection and analysis somewhat limit the generalizability of these conclusions. More precisely, limiting participating in the workshops to members of INCOSE's TLI and focusing interviews on systems engineers in leadership roles artificially restricted the different perspectives collected in the data.

The concern of potential unidentified bias in the analysis from these limitations is (somewhat) mitigated by the juxtaposition of more experienced (interviewees) with earlier career (TLI members) systems engineers. Similarly, requirements for identifying interviewees (which loosely followed the logic of snowball sampling) may not provide truly representative results for all systems engineers. The use of open-ended interview questions using terms with multiple interpretations (e.g., "failure" or "safe") and "idealistic" workshop

scenarios (e.g., less aligned with real decision-making) may have confused some participants. This kind of workshop seems a promising tool to teach system engineers about safe-to-fail probing to inform judgment. Despite these limitations, the conclusions of this study support further exploration of safe-to-fail probing.

Additionally, these conclusions offer several insights germane to system engineers. First, the data showcased how many systems engineers are "natural tinkerers" who consider failure as a normal and accepted element of innovation. While this tendency can positively impact systems design and deployment, the data highlighted that there is a distinct difference between *knowing* that it is "safe-to-fail" and *having the courage* to fail (Int. C). As previously described, there are both individual and organizational characteristics that can increase both this knowledge and this courage. Another insight relates to how safe-to-fail probes can offer clarity to systems engineers in uncertain or complex situations. Where systems engineers may naturally explore different potential pathways out of uncertainty, safe-to-fail probes remove the focus on "succeeding" and reframe it toward identifying actions to recover lost or amplify current system functionality.

Lastly, several implications stem from these conclusions and insights. First, there is a clear need for—and likely significant benefit from—deriving a clearer and more precise ontology and lexicon. Establishing a set of commonly accepted concepts, terms, and definitions will help instill safe-to-fail probes as a regular tool for systems engineers to use in informing judgement in the face of complexity and uncertainty. Second, there are richer insights to be gleaned from continuing these interviews with a broader set of participants, including from different countries, professional backgrounds, and philosophies. Here, there is an opportunity for systems engineers to identify the impact of culture differences on the interpretation of "failure" and conditions under which safe-to-fail probes are successful. Ultimately, the implication is that systems engineers can be benefit from safe-to-fail probes by eliminating "intolerable failures," tolerating "responsible failures" (Int. C) and exploiting the Tinkerer's Mindset to improve decision-making in uncertain and challenging situations.

## References


Cynefin.co (2023). *The Cynefin Framework*, https://thecynefin.co/about-us/about-cynefin-framework/

National Health Service (2021). *Safe to fail experiments.* https://www.england.nhs.uk/wp-content/uploads/2021/12/qsir-safe-to-fail-experiments.pdf. NHS England and NHS Improvement.

Kahneman, D. (2011). *Thinking Fast and Slow.* Farrar, Straus and Giroux.

Satara, Alyssa (2018), In 2 Sentences Elon Musk Explains Why the Key to Success Is Failure: If you're an entrepreneur you should follow Elon Musk's rule of thumb, *Inc. online*, https://www.inc.com/alyssa-satara/in-2-sentences-elon-musk-explains-why-key-to-success-is-failure.html

Slegers, N.J., Kadish, R.T., Payton, G.E., Thomas, J., Griffin, M.D. and Dumbacher, D. (2012). Learning from failure in systems engineering: A panel discussion, *Systems Engineering*, *15*, 74-82.

Snowden, D.J. and M.E. Boone (2007), A Leader's Framework for Decision Making, *Harvard Business Review*, *85*(11).

Snowden D. & Friends (2022). *Weaving Sense-Making into the Frabric of Our World*. Cognitive Edge.

Weiss, R. D. (1995). *Learning from Strangers: The Art and Method of Qualitative Interview Studies*. New York: The Free Press.